

Bulk stability for every monomial linearization a partial answer to arXiv:1804.11340

Candidate partial-result packet

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Status: candidate partial result likely valid. Assumptions (M1)–(M2) hold in every degree for the standard linearization of $p_k(x) = 1 + x^k$, subject to human review.

1 Source problem

Erdoes, Kruger, and Nemish prove an optimal local law for a self-adjoint polynomial once a nilpotent linearization satisfies two uniform conditions in every κ -bulk [1]. In the notation of their Assumption 2.12,

$$\begin{aligned} \text{(M1)} \quad & \sup \|M(z)\| < \infty, \\ \text{(M2)} \quad & \sup \|\mathcal{L}(z)^{-1}\| < \infty, \end{aligned}$$

where M solves the Dyson equation for the linearization and

$$\mathcal{L}[R] = R - MS[R]M.$$

They verify these conditions for general homogeneous quadratic polynomials and for symmetrized products, then ask whether an appropriate linearization works for every self-adjoint polynomial.

Assumption 2.12. There exists a constant C_3 , depending only on κ and the polynomial p , such that for any $z \in \mathbb{C}_+$ with $\operatorname{Re} z \in B_\kappa$ and $0 < \operatorname{Im} z < \infty$ we have

$$\text{(M1)} \quad \|M(z)\| \leq C_3;$$

$$\text{(M2)} \quad \|\mathcal{L}^{-1}(z)\| \leq C_3.$$

The local law is formulated using the following the notion of *stochastic domination*.

The two conditions in Assumption 2.12 of the source, page 8.

We now comment on the assumptions (M1)–(M2). We expect that these hold for an appropriate linearization for any self-adjoint polynomial, but this remains an open question in full generality. However, in Section 6.1 we prove (M1)–(M2) for a general homogeneous polynomial of degree two in Wigner matrices. For other polynomials we remark that these two assumptions can be checked numerically since they require the solution $M(z)$ of the Dyson equation for a linearization (2.7) that can be computed by an effective fixed point iteration. The numerics can be speeded up by reducing

The open question immediately after Theorem 2.14, source page 8.

2 Proof intuition

For $1 + x^k$, the source's standard symmetric linearization is a companion pencil whose coefficient of x is the reversal permutation A_k . Multiplying the pencil by A_k turns it into $xI + C_z$, where the eigenvalues of $-C_z$ are the k roots of $z - 1$. Thus the matrix Dyson solution, after the same reversal, is obtained by applying the scalar semicircle Stieltjes transform to those roots.

The vectorized stability operator has eigenvalues $1 - m_{\text{sc}}(r_i)m_{\text{sc}}(r_j)$. The semicircle identity $m^2 + rm + 1 = 0$ is rigid: a product $m_{\text{sc}}(r_i)m_{\text{sc}}(r_j) = 1$ forces $r_i = r_j$. Distinctness of the roots then reduces failure to $m_{\text{sc}}(r_i) = \pm 1$, hence $r_i = \mp 2$, which is a spectral edge. The only place where roots collide is $z = 1$, where the pushforward density is infinite. The source's κ -bulk excludes both mechanisms.

3 Partial theorem

Let A_n denote the $n \times n$ reversal permutation matrix, let $J = e_1 e_1^T$, and embed A_{k-1} in the lower-right corner by

$$D_k = \begin{pmatrix} 0 & 0 \\ 0 & A_{k-1} \end{pmatrix}.$$

Theorem 1 (All one-variable monomial degrees). *For every integer $k \geq 2$, the standard symmetric linearization*

$$L_k(x) = J - D_k + xA_k \tag{1}$$

of $p_k(x) = 1 + x^k$ satisfies assumptions (M1) and (M2) of arXiv:1804.11340 uniformly on every κ -bulk. Consequently Theorem 2.14 of the source yields the optimal bulk local law for $p_k(X^{(N)}) = I + (X^{(N)})^k$ under its Wigner moment hypotheses.

Proof. The matrix in (1) is exactly the basic monomial linearization in Appendix A.1 of the source, with constant term added in the $(1, 1)$ entry. Its Schur complement is $1 + x^k$, and the source proves that standard linearizations of this form are nilpotent.

Fix $z \in \mathbb{C}_+$, set $w = z - 1$, and define

$$C_z = ((1 - z)J - D_k)A_k.$$

Since $A_k^2 = I$,

$$(L_k(x) - zJ)A_k = xI + C_z, \quad \det(xI + C_z) = x^k - w. \tag{2}$$

For $w \neq 0$, let $r_1, \dots, r_k$ be the distinct roots of $r^k = w$. Equation (2) says that C_z is diagonalizable with eigenvalues $-r_j$.

Let s be a standard semicircular element with law μ_{sc} . The source's construction of the Dyson solution gives

$$M(z) = \int_{-2}^2 (L_k(x) - zJ)^{-1} d\mu_{\text{sc}}(x).$$

Writing

$$m_{\text{sc}}(r) := \int_{-2}^2 \frac{d\mu_{\text{sc}}(x)}{x - r},$$

functional calculus in (2) shows that the eigenvalues of $M(z)A_k$ are

$$m_j = m_{\text{sc}}(r_j), \quad j = 1, \dots, k. \tag{3}$$

The chosen semicircle branch satisfies

$$m_{\text{sc}}(r)^2 + r m_{\text{sc}}(r) + 1 = 0, \quad r = -(m_{\text{sc}}(r) + m_{\text{sc}}(r)^{-1}). \quad (4)$$

We first establish (M2). In the present one-variable pencil the self-energy is $\mathcal{S}[R] = A_k R A_k$. Both A_k and M are symmetric. With column vectorization,

$$\text{vec}(M A_k R A_k M) = ((A_k M)^T \otimes M A_k) \text{vec } R = (M A_k \otimes M A_k) \text{vec } R.$$

Therefore $\mathcal{L}$ is singular precisely when

$$m_i m_j = 1 \quad (5)$$

for some i, j . If (5) holds, then (4) gives

$$r_j = -(m_j + m_j^{-1}) = -(m_i^{-1} + m_i) = r_i.$$

When $w \neq 0$ the roots are distinct, so $i = j$. Then $m_i^2 = 1$, and (4) yields $r_i = \mp 2$. Hence $w = r_i^k = (\mp 2)^k$, a spectral-edge value. For $\Im z > 0$ this is impossible, and for a real boundary value it is outside every κ -bulk.

It remains to justify uniformity and (M1). The limiting density of $p_k(s) = 1 + s^k$ is the elementary pushforward

$$\rho_k(E) = \sum_{\substack{x \in (-2, 2) \\ 1 + x^k = E}} \frac{\sqrt{4 - x^2}}{2\pi k |x|^{k-1}}. \quad (6)$$

Thus a fixed κ -bulk stays a positive distance from the edges and from $E = 1$, where the density diverges. In particular, $|w|$ is bounded below and the roots r_j remain uniformly separated when $0 \leq \Im z \leq 1$ and $\Re z$ lies in that bulk. Diagonalizing C_z in (2), the boundary values in (3) are uniformly bounded, and so is $M(z)$. This proves (M1) for $0 < \Im z \leq 1$.

On the same compact parameter set, the preceding product argument shows that $\mathcal{L}(z)$ is never singular, including at the boundary. Continuity and compactness therefore give a uniform bound for $\|\mathcal{L}(z)^{-1}\|$. For $\Im z \geq 1$, the source's general nilpotent-resolvent estimate bounds M . Moreover, direct adjugate expansion of (2) shows that $M(z)A_k$ has a finite limit as $\Im z \rightarrow \infty$: the denominator is $x^k - w$, while every cofactor is affine in the rank-one parameter w . All its eigenvalues $m_{\text{sc}}(r_j)$ tend to zero, so this limiting matrix is nilpotent. Hence $I - M A_k \otimes M A_k$ has an invertible limit. Enlarging the constant over the remaining compact strip completes both uniform bounds. Thus (M1)–(M2) hold exactly in the sense of Assumption 2.12. $\square$

4 Verification report

The proof was checked in four independent ways.

1. Direct symbolic inversion gives $(L_k(x) - zJ)_{11}^{-1} = (1 + x^k - z)^{-1}$.
2. For $2 \leq k \leq 12$, the included SymPy check confirms $\det((L_k(x) - zJ)A_k) = x^k - z + 1$ exactly.
3. The vectorization identity reduces the k^2 -dimensional stability determinant to the pairwise products in (5).
4. Numerical root checks for degrees 3, 4, 7, 10 at three nonreal spectral parameters confirm the semicircle quadratic identity and find no unit product.

The computational checks do not prove the all-degree theorem; the companion factorization and identity (4) do.

5 Scope and limitations

This proves the source’s desired stability for the infinite family of one-variable monomials, but not for arbitrary sums or genuinely noncommutative multivariable polynomials. The remaining difficulty is that a general linearization need not reduce to a single companion matrix whose stability eigenvalues are scalar semicircle transforms of distinct roots.

For a single Wigner matrix, the local law for $1 + X^k$ can also be transferred from the usual Wigner local law by spectral mapping away from critical values. The contribution here is therefore the exact verification of the source’s linearization conditions (M1)–(M2), not a previously inaccessible one-variable universality phenomenon.

6 Bounded novelty check

The search through 11 August 2026 covered the run registry and packet indexes; the exact open sentence; arXiv:1804.11340 and its title combined with *general polynomial*, *monomial*, *minimal linearization*, *stability operator*, and *bulk local law*; and later arXiv work on polynomials of Wigner matrices. The search found later edge results for multivariate quadratic polynomials, but no exact all-degree monomial verification of (M1)–(M2). Novelty confidence is moderate-to-low because the companion-matrix mechanism may be known folklore; mathematical validity does not depend on novelty.

7 Conclusion and review recommendation

The full arbitrary-polynomial question remains open, but every monomial degree admits the desired stable linearization. Promotion as a likely valid partial result is recommended after checking the compactness argument and the source’s linearization sign convention.

References

- [1] L. Erdos, T. Kruger, and Y. Nemish, *Local laws for polynomials of Wigner matrices*, J. Funct. Anal. 278 (2020), 108507; arXiv:1804.11340. <https://arxiv.org/abs/1804.11340>.